

them into overarching symbolic reasoning. As concerns the complexity of different kinds of reasoning, reasoning about how to interpret perceptual inputs (carried out during Perception Interpretation) is every bit as difficult as reasoning about what to do next (carried out during Deliberation).

Calling this module *Deliberation* rather than *Reasoning* aims to preclude the reader from automatically making inferences about the nature of this module in the LEIA's architecture. However, it would be unnatural to avoid the term *reasoning* altogether. So, we trust that readers will take to heart the pervasiveness of reasoning throughout LEIA operation and will readily understand the context-specific implications of each use of the term *reasoning*.

LEIA reasoning emulates that of people in important ways. First, it takes into account beliefs, biases, intuitions, decision-making heuristics, and other less-than-perfectly-rational facets of human functioning.¹³ Second, it orients around actionability. This means that LEIAs are designed to detect when they have understood a situation well enough to act. In high-risk situations, the bar for actionability will be very high, whereas in low-risk situations, the agent can chance a misstep, which can subsequently be corrected through interaction with its human partner. Finally, LEIA reasoning deemphasizes reasoning from first principles, which was a main focus of early AI. As a consequence, the efficiency of search is not a central concern of LEIAs. Instead, they pursue their goals using stored plans. Although people are certainly capable of search, they seem to use it in rather limited contexts, outside of game playing. Accordingly, content-centric modeling emphasizes the use of remembered, habitual solutions for heuristic decision-making. When no known plan leads to a goal, LEIAs, as social agents, prefer to ask for help rather than explore independently. This might be viewed as a facet of the cognitive miser theory (Stanovich, 2009), which posits that the *principle of least effort* applies to cognitive effort no less than to other human endeavors. Preferring communication to exploration in order to economize effort is only viable for language-endowed social agents. If collaborating with a human is, for some reason, not possible at a given moment, then the agent may opt to engage in reasoning from first principles, as by modifying a known plan using partial matching and reasoning by analogy. For example, remembering how lenses can be used to start a fire with sunlight allowed the modern Robinson Crusoe, Alvaro Cerezo, to use a plastic bag filled with sea water for this purpose.¹⁴ He did not

need to build a plan of action from scratch; he just amended a known plan by fulfilling a prerequisite—the availability of a lens—in a novel way. He could do this because of his understanding of the nature of lenses. In sum, content-centric modeling strives to replace search by retrieval from structured knowledge. It underscores the centrality of heuristics and downplays the dynamic generation of search spaces and search as such.

The kinds of reasoning undertaken in the Deliberation module are best explained using examples, which are amply provided in chapters 6–8. For comparison, reasoning in data-driven AI is discussed in section 2.7.1.

2.1.4 Action Specification

The output of the Deliberation module is the agent's decision about what to do, but not yet how to do it. How to do it is determined in the Action Specification module. For example, when asked a question, the agent must decide how many and which details to provide in the answer, which mode of communication to use (language and/or body language), which stylistic features to express (politeness, formality, enthusiasm), and which actual sentences and/or gestures will best serve in the given context. All of these aspects of Action Specification will be illustrated in upcoming chapters using examples of dialog interactions. The output of Action Specification is a meaning representation that needs to be rendered as an actual action using the appropriate effector, such as the text generator, the speech synthesizer, or robotic movement effectors.

2.1.5 Action Rendering

Action rendering is most naturally implemented using data-driven methods developed in fields such as robotics, video simulation, and text-to-speech generation. As such, action rendering lies outside of a LEIA's cognitively modeled core. However, the results of cognitive processing must be translated into the formats expected by data-driven action generation modules, just as the results of data-driven processing must be cognitively interpreted at the Perception Interpretation stage. For natural language generation, Action Rendering is divided into multiple stages that use different data-driven tools with different rationales and expectations of their performance (cf. section 4.3); this is a good example of how symbolic and data-driven approaches are being strategically integrated in LEIAs.